Home Reading 03 — Transfer Functions

Before Class

Preview Lecture 3

Read Lecture 3: Transfer Functions.

Textbook Reading

- Required: Ogata, System Dynamics, 4th ed., Sections 4-1–4-2, pp. 106–110 (transfer functions and block diagrams).
- Page numbers above are printed textbook pages. Use the Lecture slides for the remaining worked examples.

This Week Introduces

Zero-state transfer functions, block diagrams, poles and zeros, and the response of one model to different inputs.

Tutorial Preparation

Read Tutorial 3 before class. Be ready to:

- Q1: Check the gain, poles and selected input-output pair.
- Q2: Write the massless-node balance before eliminating the internal variable.
- Q3: Trace the disturbance entry point and summing signs in a block diagram.
- Q4: Separate initial-state free motion from a transfer-function response.

Check Your Understanding Before Class

Before class, try to answer these two questions:

- Why must a transfer function state both its input and its output?
- Why are zero initial conditions part of the transfer-function definition?

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